

ASTRON 1120: Stars, Stellar Structure, and Stellar Evolution

Final Project

1 Basic Rules

The final project is worth 40% of your final ASTRON 1120 grade, 30% for the written report, and 10% for the oral presentation. The main goal of the final project is to give you the opportunity to work on a real problem in stellar astrophysics and present the results in an academic setting. By design, the scope of the final project is not completely defined; it is up to each group to decide how much they want to do (and of course, the more you do, the better your grade will be). You can either propose a topic of your own (which I will value very positively), or choose among the topics that I suggest below.

Groups The final project should be completed in groups of **two or three people**.

What to aim for Ideally, you should do some kind of non-trivial exercise involving either stellar models (i.e., MESA calculations or similar), astronomical data sets (photometry, spectroscopy, astrometry, etc.), or a combination of models and data, related to the topics covered in ASTRON 1120. The suggestions below will give you an idea of what to aim for, but feel free to surprise me with something new and original: I will value this very positively! The bare minimum effort should be roughly equivalent to four times the work required to complete one of the 10-point ASTRON 1120 Assignments, but the more you do, the better your grade will be.

Reports The main deliverable from this exercise (30% of the final course grade) will be a written report due **Monday, April 20** (the Monday of the last week of classes). The reports will have a **Strict length limit of four pages**, and must include an abstract, introduction, conclusions, and references. The reports might have other sections, but these four are *mandatory*. I expect every report to include a few plots, which should be of high enough quality to include in a scientific paper. I will leave the details of the report format to you, but if you want to make it look good, I suggest you use the Overleaf template for AAS journals, which you can find in [\[this link\]](#). Together with the report, you should submit an iPython notebook and data files that can reproduce your main plots and results. I will not grade the code itself, but I will check that it works.

Presentations Each group will present their results in class during the last week of classes, on **April 21 and 23**. The presentation will be worth 10% of the final course grade. All presentations will have a **strict time limit of 15 minutes**. I will be ruthless with this, so do not go over time! I expect that most of you will use Power Point, Keynote, or some other presentation software to produce slides, but you can also do derivations on the board, show short movies, or use any other tool that you think will help, as long as you stay within your time limit. After each presentation, there will be 4 minutes set aside for questions and discussion. I will definitely ask questions after each presentation, and I will expect the other students to ask questions of their own, and engage in the discussion.

Logistics Each group should send me an email stating their members and their chosen project topic on or before **Thursday, March 5** - the Thursday before Spring Break. To make sure that every group is on track to deliver something of substance by the April 20 deadline, I will expect a milestone report on or before **Thursday, March 26**. This report can be as brief as an email with a couple of sentences and a plot or two. It is completely acceptable to say that you are stuck trying something or that you don't have any results yet, but the milestone report should show that you have at least started working on the project and that you have made some kind of progress. I encourage each group to come to office hours to ask questions about their projects, but please do this as a group (i.e., talk within the group before coming to my office) and please use Google and Stack Overflow before deciding that you really are stuck and need help.

Grading I will grade your final project based on the amount and quality of the work you have done, the clarity of the report, and the quality of the plots. I will grade the oral presentations based on the clarity of the presentation, the attention to context, and the effectiveness with which you convey the work reflected in the report.

2 Proposed Project Topics

These topics are roughly ordered from easiest to most challenging.

The mass-radius plot as a function of metallicity: In [Webbink \(2008\)](#), Ron Webbink draws an interesting plot (his Figure 1), mapping the radius of stars at different key moments in their evolution as a function of mass. As explained in the paper, the radius and mass of a star determine when it will fill its Roche lobe and start interacting with a potential companion. Your assignment is to do a complete overhaul of the mass-radius plot using modern stellar evolution models (MIST or equivalent) and taking into account the effect of metallicity. Metallicity matters because recent work by eminent researchers has found a strong anti-correlation between stellar metallicity and the close binary fraction; see [Moe et al. \(2019\)](#) (this is a *very long* article - you might want to focus on Figures 18 and 19, and the accompanying discussion). Based on these findings and your revision of the mass-radius plot, we make broad predictions for the rate of binary interactions in stellar populations of different metallicities.

The Initial-Final Mass Relation as a function of metallicity: Using MIST models, or another set of publicly available stellar evolution tracks, explore the relation between a star's initial (ZAMS) mass and the mass of the WD that it forms. Note how this depends on metallicity. Search the literature for measurements of the initial-final mass relation, and compare them with the predictions from the MIST models.

Ages of globular clusters in Gaia: In A5, you developed the tools to fit for the distance modulus and stellar age of stellar populations in globular clusters using MIST isochrones. [Vasiliev & Baumgardt \(2021\)](#) describe a method to identify globular cluster members in Gaia EDR3 data, which they use to calculate mean parallaxes for many individual clusters and perform a series of quality tests and benchmarks on the Gaia data. Your assignment is to measure the ages of at least ten of the clusters studied by [Vasiliev & Baumgardt \(2021\)](#) using the MIST isochrones and the superb photometry from Gaia. Conveniently, [Vasiliev & Baumgardt \(2021\)](#) provide a link to a Zenodo repository that lists all Gaia sources in the field of

each cluster with their membership probabilities, and all the data that you need to produce color magnitude diagrams of each cluster.

The Transition to Fully Convective Stars in Gaia: Jao et al. (2018) show a gap in the lower main sequence around an absolute magnitude of 10 in the open *G* filter used by Gaia, and identify this gap with the transition to fully convective stars at the low mass end of the main sequence. Your assignment is to figure out how to download data from the [Gaia Archive](#), produce a catalog of objects with enough statistics to reproduce Figure 1 in Jao et al. (2018), and compare these data to models to test the hypothesis that the transition to fully convective stars is indeed responsible for the gap.

Bursts in MESA Go to [this website](#). This is one of many on-line tutorials that use MESA, but it might be the most complete. Follow the step-by-step instructions to download and install MESA, execute a few basic runs, and simulate X-ray bursts and recurrent novae. If you choose this option, I expect you to complete the last section (on recurrent novae), and run enough simulations to make a crude map of the stability boundary.

3 Accessing the Technical Literature

Most of the references you will need for any astrophysics-related project can be found on the NASA Astrophysics Data System, or ADS, at <https://ui.adsabs.harvard.edu/>. A few search queries should quickly get you to the reference you are looking for. The ADS usually lists authors, title, and abstract, and a link to the actual publication. Unfortunately, some of the leading journals in astrophysics are behind a paywall, but there is usually a way to access the pdfs free of charge. If you execute your query within the Pitt web domain, that should give you access to many of the main journals, but in case that does not do the trick, remember most authors also post their publications on the <https://arxiv.org/> web sever (which ADS links to by default), and those are always free. Let me know if you have any problems accessing the publications you need to complete your project.

4 Some Tips on Technical Writing

- Be brief and to the point. No anecdotes or detective stories - just describe what you did and how you did it.
- However, do not be obscure. Explain things as clearly as you can. Someone reading your report should be able to reproduce what you did, without guesswork.
- Be organized. Decide what you are going to present (plots, tables) *before* writing, and build your text around it. Needless to say, this will only work if you finish the project with enough time to write the report.
- Remember to properly reference your sources. This is not limited to scholarly articles: software packages and online data repositories should also be referenced.

5 Some Tips on Technical Presentations

Making effective technical presentations is not easy, but it is an extremely useful skill. Practice and preparation can go a long way, so take this exercise as an opportunity to improve your presentation skills. As a rule of thumb, you should try to devote 1/3 of your time to explaining the context and importance of the problem, 1/3 to the main methods used to study the problem, and 1/3 to conclusions, clearly explaining how the problem fits in the general context, why it is useful and important, and what are the main open questions left to answer. Note that context is crucial; unless you can convince the audience that what you are going to tell them is meaningful, they are not going to care about any of the details. A common mistake that people make is focusing on the part of the topic that was the hardest to understand or that took them the longer to figure out. You should try to avoid that, especially if it compromises the flow of the narrative and does not let you present the context adequately. There is usually a chance to give more detail on the technical aspects of a topic during the question time. And never, ever, go over time. I strongly recommend that you do a few practice runs and time them to make sure this does not happen.

There are many websites and manuals on how to make good viewgraphs for technical presentations and how to give good talks. There are a few common sense rules, but many things are a matter of personal choice. Therefore, I will give you a lot of freedom on how to structure your talk. If you need further guidance, I suggest you read these short articles:

- *How to Give a Good Talk* by Uri Alon [\[link\]](#).
- *10 Tips for Academic Talks* by Matt Might [\[link\]](#). You might want to check the rest of Matt's website too, it is full of useful stuff.
- *Giving an Academic Talk* by Jonathan Shewchuk [\[link\]](#)

One final word about questions and discussion. The nature of this exercise involves a certain degree of scrutiny, critique, and feedback on your presentation. That should not be a problem. We should all work to ensure that this is done in a relaxed, laid-back, non-confrontational manner. Everybody is allowed to ask questions, raise issues, and make observations about all presentations, but this should always be done in a respectful manner. I will allow no exceptions to this rule.

References

- Jao, W.-C., Henry, T. J., Gies, D. R., & Hambly, N. C. 2018, [??jnlApJ](#), 861, L11
- Moe, M., Kratter, K. M., & Badenes, C. 2019, [??jnlApJ](#), 875, 61
- Vasiliev, E. & Baumgardt, H. 2021, [??jnlMNRAS](#), 505, 5978
- Webbink, R. F. 2008, in *Astrophysics and Space Science Library*, Vol. 352, *Astrophysics and Space Science Library*, ed. E. F. Milone, D. A. Leahy, & D. W. Hobill, 233